

Thomas Ngo

Email: duchoang@ualberta.ca

Location: Edmonton, AB

Phone: (825) 712-7485

LinkedIn: [linkedin.com/in/thomas-ngo-280539327](https://www.linkedin.com/in/thomas-ngo-280539327)

Portfolio: dev-portfolio-alpha-lovat.vercel.app

GitHub: github.com/chauu335

EDUCATION

Bachelor of Science: Computing Science (Honors)

Edmonton, AB

University of Alberta

Sep 2023 – Expected June 2027

- **GPA: 3.9 / 4.0**
- **Honors & Awards:** International Admission Scholarship, Regional Excellence Scholarship.
- **Relevant Coursework:** Machine Learning I & II, Algorithms, Visual Recognition (Computer Vision), Software Engineering, Web Applications & Architecture, Natural Language Processing, Mathematical Statistics, Numerical Methods.

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, TypeScript, C/C++, C#, SQL, R, LaTeX, HTML/CSS.
- **AI & Data Science:** Deep Learning, Natural Language Processing, Computer Vision (OpenCV), LLM Integration, Scikit-Learn, PyTorch, NumPy, Pandas, Time Series Analysis, Google Gemini API, Claude API.
- **Frameworks & Web:** React, Node.js, Express, Spring Boot, Tailwind CSS, JUnit, RabbitMQ, Playwright, Maven.
- **Cloud & DevOps:** Docker, Git, CI/CD Pipelines, Firebase, Linux/Unix.
- **Databases:** PostgreSQL, MySQL, MongoDB, SQLite3, Firestore.

PROFESSIONAL EXPERIENCE

Software Engineering Intern

Remote / Hanoi, Vietnam

Viettel Military Industry and Telecoms Group

May 2026 – July 2026

- Designed and delivered the **Agri Rescue Platform** (Java, Spring Boot, React, TypeScript, PostgreSQL, RabbitMQ, Docker), a full-stack microservices platform for surplus agricultural produce rescue, after completing enterprise architecture training. Applied **Agile/Scrum** practices throughout development and defended the final project to company leadership.
- Designed a modular **REST API** across **7 microservices** with **JWT-based authentication** and an **event-driven** architecture (**RabbitMQ** + outbox pattern), using **pessimistic/optimistic locking** and a **finite state machine** to enforce a valid produce lifecycle behind a **JWT-secured Spring Cloud Gateway**.
- Built a **React/TypeScript** frontend dashboard with an **LLM-powered** analytics assistant (Google Gemini API), and containerized the full system with **Docker Compose** over a **schema-per-service PostgreSQL** architecture.
- **GitHub:** github.com/chauu335/agri-rescue-platform

Software & Digital Operations Intern

Calgary, Canada

Cosmos Collision Ltd

May 2026 – August 2026

- Authored standard operating procedures (SOP) and managed daily operations for 3D scanning, 3D printing, and CNC carving equipment to speed up internal fabrication workflows.
- Built a Chrome extension that auto-extracts data from collision estimate PDFs and fills out Mitchell intake forms, cutting manual data entry across contact/vehicle/insurance sections and reducing transcription errors.

SELECTED PROJECTS

Android Event Lottery Application

Oct 2025 – Dec 2025

Java, Firebase, CI/CD | **GitHub:** github.com/CMPUT301F25basket/shopping-basket

- Co-developed a native Android application using **Java** and **Android SDK**, building modular UI layouts with **XML** and **Fragments** for smooth navigation across 500+ prospective users.
- Implemented real-time data synchronization using **Firebase Firestore** (NoSQL) for live event tracking, participant lottery pools, and instant **QR-code** check-in validation.
- Engineered Role-Based Access Control (**RBAC**) workflows for admin, organizer, and participant roles, integrating dynamic push notifications and automated lottery selection algorithms.
- Configured automated **CI/CD pipelines** with **JUnit** unit testing, ensuring zero-regression code updates and reducing deployment overhead by **30%**.

Optimizer Animated Visualizer

May 2026

React, TypeScript, Python, Node.js, Railway | **GitHub:** github.com/chauu335/optimizer-visualizer

- Developed a full-stack benchmarking platform to simulate and visualize 12 ML optimizer-kernel combinations; engineered a **Python/NumPy** engine to execute high-performance numerical optimization experiments.
- Built an interactive **React/TypeScript** dashboard featuring real-time playback controls and dynamic animations using **Recharts**, enabling step-by-step analysis of learning curve convergence.
- Designed a modular **RESTful API** to bridge the numerical backend with the frontend, deploying the system via **Railway** with an automated CI/CD pipeline for fast iteration.

EXTRACURRICULAR & LEADERSHIP

VP of Marketing

May 2025 – April 2026

UAlberta Book Club

- Led multi-platform marketing strategies and social media campaigns across Instagram and campus channels to grow visibility and active membership.
- Designed graphics, promotional assets, and event announcements, increasing member turnout and club engagement.
- Collaborated with executive team members to plan and promote large-scale events, book drives, and community meetups.

Student Volunteer

Oct 2024

UofA Computing Science Department

- Represented the Department of Computing Science at prospective student events, engaging with hundreds of visitors to provide specific guidance on degree paths.
- Addressed inquiries regarding undergraduate course structures, faculty resources, research opportunities, and student life to assist prospective students in their academic decision-making.